VISION FURNISH: AUGMENTED REALITY FURNITURE APPLICATION

A Minor Project Report

Submitted in partial fulfillment of requirement of the

Degree of

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING

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Report Approval

The project work **"Vision Furnish: Augmented Reality Furniture Application"** is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion drawn there in; but approve the "Project Report" only for the purpose for which it has been submitted.

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Declaration

I/We hereby declare that the project entitled **"Vision Furnish: Augmented Reality Furniture Application"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology in '**Department of Computer Science and Technology**' completed under the supervision of **Prof. Vivek Kumar Gupta** and **Prof. Diptanshu Pandya,** Faculty of Engineering, Medi-Caps University Indore is an authentic work.

Further, I/we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

Signature and name of the student(s) with date

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Certificate

I/We, **Prof. Vivek Kumar Gupta** and **Prof. Diptanshu Pandya** certify that the project entitled **"Vision Furnish: Augmented Reality Furniture Application"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications by **Manya Arora, Mariya Mirza, Nikita Nair** is the record carried out by him/them under my/our guidance and that the work has not formed the basis of award of any other degree elsewhere.

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Abstract

The rapid evolution of e-commerce has fundamentally changed consumer behavior, yet the furniture industry faces a persistent hurdle: the "spatial visualization gap." Consumers often struggle to judge how a product's dimensions, aesthetics, and color palette will harmonize with their existing home environment. **Vision Furnish** is an Augmented Reality (AR) application designed to bridge this gap. Utilizing state-of-the-art **Simultaneous Localization and Mapping (SLAM)** technology, the application allows users to project hyper-realistic, 3D digital twins of furniture into their physical living spaces using only a smartphone.

This project explores the integration of **Google's ARCore** and **Unity 3D** to create a seamless user experience that includes real-time plane detection, light estimation, and physics-based scaling. The results of the implementation indicate a significant improvement in user confidence and a reduction in the "misfit" purchase rate. By providing a "try-before-you-buy" digital experience, Vision Furnish represents the future of interactive retail, making interior design accessible, accurate, and cost-effective.

Keywords:

- Augmented Reality (AR)
- Vision Furnish
- Furniture Visualization
- ARCore
- Unity 3D
- Simultaneous Localization and Mapping (SLAM)
- Plane Detection
- 3D Model Placement
- Interior Design
- Real-Time Rendering
- Spatial Visualization
- Markerless AR
- Firebase
- Mobile Application
- User Experience (UI/UX)
- Light Estimation
- Object Scaling
- Digital Twin
- Android Application
- Smart Retail

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Abbreviations

Abbreviation	Full Form
AR	Augmented Reality
VR	Virtual Reality
MR	Mixed Reality
SLAM	Simultaneous Localization and Mapping
UI	User Interface
UX	User Experience
SDK	Software Development Kit
API	Application Programming Interface
IMU	Inertial Measurement Unit
CAD	Computer-Aided Design
PBR	Physically Based Rendering
DFD	Data Flow Diagram
ER	Entity Relationship
IDE	Integrated Development Environment
BaaS	Backend as a Service
FPS	Frames Per Second
GC	Garbage Collection
OIS	Optical Image Stabilization
DOF	Degrees of Freedom
CDN	Content Delivery Network
JWT	JSON Web Token
APK	Android Package Kit
NDK	Native Development Kit
NoSQL	Not Only SQL
URL	Uniform Resource Locator

Chapter 1

Introduction

1.1 Introduction

In the modern digital era, Augmented Reality (AR) has transitioned from a futuristic concept to a practical tool for solving real-world consumer problems. Interior design is a domain that relies heavily on spatial awareness and visual harmony.

Traditionally, consumers relied on physical measurements and imagination, which are prone to human error. **Vision Furnish** addresses these challenges by creating a middle ground between the digital storefront and the physical home.

The application leverages the camera and sensors of a mobile device to "read" the environment. It identifies surfaces like floors and walls, allowing users to place virtual furniture that behaves as if it were physically present. Through advanced rendering techniques, the app ensures that the virtual objects maintain their position and scale even as the user moves around the room, providing a 360-degree perspective of the potential purchase.

1.2 Literature Review

The evolution of AR in interior design can be categorized into three distinct eras:

1. **The Marker Era:** Early applications required a physical printed marker (like a QR code) to be placed on the floor to serve as an anchor. This was cumbersome and limited user adoption.
2. **The Depth-Sensing Era:** Devices equipped with specialized hardware (like Microsoft HoloLens or early Project Tango phones) used infrared sensors to map rooms. While accurate, the hardware was too expensive for the average consumer.
3. **The Modern SLAM Era:** Current frameworks like **ARCore (Google)** and **ARKit (Apple)** use **Simultaneous Localization and Mapping (SLAM)**. This technique uses the standard smartphone camera to track "feature points" (corners of rugs, patterns on a floor) and combines this with **Inertial Measurement Unit (IMU)** data (gyroscope/accelerometer) to build a 3D map of the environment in real-time. Vision Furnish leverages this third era to provide high-end visualization on consumer-grade hardware.

1.3 Objectives

The primary objectives of this minor project are:

1 **To Develop a Markerless AR Environment:** Utilize ARCore/ARKit to detect horizontal planes (floors) and vertical planes (walls) with high precision.
2 **To Implement Real-time 3D Asset Integration:** Enable the fetching and rendering of high-polygon 3D models $(\text{.glb/ .fbx formats})$ without significant frame-rate drops.
3 **To Ensure Spatial Accuracy:** Implement 1:1 scaling so that a virtual 6-foot sofa occupies exactly 6 feet of digital space in the AR viewport.
4 **To Enhance User Interaction:** Develop intuitive UI/UX for rotating, moving, and swapping textures of the furniture items.
5 **To Optimize Performance:** Ensure the application remains stable on mid-tier hardware through mesh decimation and texture compression.

5.5 Significance

The significance of Vision Furnish lies in its dual benefit to consumers and retailers:

- **For Consumers:** It eliminates the manual labor of measuring rooms and the psychological stress of "buyer's remorse." It democratizes interior design, allowing anyone with a smartphone to visualize professional-grade layouts.
- **For Retailers:** It reduces the logistics costs associated with product returns. In the furniture industry, returns due to "size mismatch" are expensive; AR virtually eliminates this issue.
- **Technological Contribution:** The project demonstrates the practical application of computer vision and mobile sensor fusion in a commercial context.

1.5 Research Design

The project utilizes an **Agile Developmental Research Design**, characterized by the following phases:

- **Environmental Analysis:** Studying how different floor textures (marble, wood, carpet) and lighting intensities (incandescent vs. natural sunlight) affect the stability of the AR anchors.
- **Asset Optimization Pipeline:** A critical research component involved determining the balance between **Polygon Count** and **Mobile Performance**. High-detail models (100k+

polygons) cause lag, while low-detail models look unrealistic. The project settled on a **5,000 to 15,000 polygon limit** per model using **PBR (Physically Based Rendering)** textures to maintain realism.

- **Quantitative Validation:** The research includes a "Ground Truth" comparison. A physical object of known dimensions is measured, and its digital twin is placed in AR. The deviation (error margin) is recorded to ensure the scaling algorithm is accurate within a **±1% range**.

1.6 Source of Data

- **Spatial Data:** Real-time point cloud data generated by the ARCore/ARKit SDK during the scanning phase.
- **3D Geometric Data:** CAD models and .glb/.gltf files sourced from architectural repositories, ensuring all models adhere to standard manufacturing dimensions (e.g., standard dining table height of 30 inches).
- **User Feedback Data (Qualitative):** Preliminary testing with a control group to identify "UI Friction"—observing where users struggle with the gesture controls (rotate/scale).

1.7 Chapter Scheme

The report is organized into multiple chapters covering system requirements, design architecture, implementation methodology, testing procedures, results analysis, conclusions, and future scope.

Chapter 2 presents the Requirement Specification.
Chapter 3 discusses the System Design and architecture diagrams.
Chapter 4 covers Implementation, Testing, and Maintenance.
Chapter 5 presents Results and Discussion.
Chapter 6 provides Summary and Conclusions. Chapter
7 outlines Future Scope.

Chapter 2

Requirement specifications

2.1 User Characteristics

The application is designed for a broad spectrum of users, categorized as follows:

- **The Homeowner/End-User:** Individuals seeking to renovate or purchase furniture. They require a frictionless, high-speed interface where the complexity of AR mapping is hidden behind a "point-and-click" experience.
- **The Interior Design Student:** Users who may use the app for academic layouts. They require high accuracy in object scaling and realistic light estimation.
- **The Administrator (Retailer):** Responsible for managing the digital catalog. They interact with the system via a web-based backend (e.g., Firebase Console) to upload new .glb models and update pricing or descriptions.

2.2 Functional Requirements

A. Environmental Analysis Module

- **Feature Point Mapping:** The system must identify at least 50 unique feature points in a $3m \times 3m$ area to establish a stable tracking coordinate system.
- **Instant Placement (Depth API):** On supported Android devices, the app should use the **ARCore Depth API** to place objects instantly without waiting for a full plane scan.
- **Vertical Plane Detection:** The system must detect walls to allow for "Wall-mounted" furniture like paintings, shelves, or mirrors.

B. Interaction & Manipulation Module

- **Haptic Feedback:** The Android device should provide a short vibration (Haptic) when a 3D object successfully "snaps" onto a detected floor plane.
- **Collision Detection:** The system should prevent two virtual objects from overlapping (Inter-object collision) to maintain realism.
- **Bounding Box Visualization:** When an object is selected, a thin wireframe box should appear to indicate its selection and physical boundaries.

C. Content Management

- **Lazy Loading:** 3D models should not load all at once. The app must implement "Lazy Loading" to download only the model the user selects, saving mobile data.
- **Cache Management:** Downloaded furniture models must be cached locally in the Internal Storage/Android/data/ folder so they can be used offline later.

2.3 Dependencies

Component	Technology	Reason for Use
Core AR Engine	**ARCore SDK v1.40+**	Industry standard for Android; provides SLAM and Light Estimation.
Rendering Engine	**SceneView / Filament**	Google's physically-based rendering (PBR) engine for realistic textures.
Language	**Kotlin 1.9+**	Modern Android standard; null-safety is crucial for high-performance AR.
UI Framework	**Jetpack Compose**	To build a modern, overlay-based UI that sits on top of the camera view.
Cloud Integration	**Firebase Firebase/Firestore**	To handle real-time catalog updates and high-speed asset delivery.

2.4 Performance Requirements

To provide a professional experience, the app must meet these performance benchmarks:

- **Frame Rate Stability:** The AR session must run at a consistent **30-60 FPS**. If the frame rate drops below 24 FPS, tracking becomes jittery, causing "drift."
- **Initialization Speed:** Surface detection (finding the first plane) should occur within **5–10 seconds** of moving the camera.
- **Memory Management:** The application must utilize **Garbage Collection (GC)** optimization to keep the heap size under **250MB** even when rendering high-poly models.
- **Network Efficiency:** High-quality furniture models should be compressed into .glb format to ensure the download size per model does not exceed **10MB**

2.5 Hardware Requirements

As an AR application, the hardware requirements are more stringent than standard Android apps:

- **Processor:** Minimum **Octa-core 2.0 GHz** (Snapdragon 700 series or equivalent).
- **RAM:** Minimum **4GB** (6GB+ is recommended for stable 3D rendering).
- **Camera:** A high-resolution camera with **Auto-Focus** and **Optical Image Stabilization (OIS)** if possible.
- **Sensors: Gyroscope, Accelerometer, and Magnetometer** are mandatory for tracking the device's 6-DOF (Degrees of Freedom).
- **Storage:** At least **500MB** of free space for caching 3D models.
- **Target Devices:** Devices on the **Google Play Services for AR** supported list (e.g., Pixel 3+, Samsung S10+, etc.).

2.6 Constraints & Assumptions

Constraints

- **API Level:** The app is constrained to **Android 8.0 (Oreo) - API Level 26** or higher because ARCore features are unstable on older versions.
- **Processing Power:** The app may experience thermal throttling on budget Android devices during extended 3D rendering sessions.
- **Asset Complexity:** Total polygon count in a single "scene" must stay below **100,000 polygons** to avoid crashing the GPU.

Assumptions

- **Environmental Lighting:** It is assumed the user is in a room with at least **100 lux** of light. ARCore cannot track feature points in pitch-black or dimly lit environments.
- **Surface Texture:** It is assumed the floor is not a "perfect mirror" or single-color surface. SLAM technology requires visible texture (like wood grain or carpet patterns) to track movement.
- **User Movement:** It is assumed the user will move the device in a slow, sweeping motion during the "Scanning" phase as prompted by the UI.

Chapter 3

System Design

3.1 Algorithm: Anchor-Based Object Placement (SLAM)

The core logic follows the **Concurrent Odometry and Mapping** approach.

1. **Start:** Initialize ArSession and check for AR_CORE_INSTALL_STATUS.
2. **Feature Extraction:** Capture camera frames and identify high-contrast "Feature Points."
3. **Pose Estimation:** Compare feature point displacement with **IMU (Gyro/Accel)** data to calculate the device's 6-DOF (Degrees of Freedom) pose.
4. **Plane Detection:** Perform **Hit-Testing**. If a cluster of feature points remains coplanar, define an ArPlane.
5. **Anchor Creation:** When the user taps a screen coordinate (x, y), cast a ray. If the ray intersects an ArPlane, create an ArAnchor.
6. **Rendering:** Instantiate the .glb model from the ModelRenderable library and attach its transform to the ArAnchor.
7. **Update Loop:** Continuously recalculate the model's position relative to the camera to ensure it stays "pinned" to the real world.

3.2 Function-Oriented Design (Modules)

In a procedural context, the system is divided into functional modules:

- **Initialization Module:** Handles camera permissions and ARCore compatibility checks.
- **Tracking Module:** Manages the camera stream and point-cloud generation.
- **Asset Management Module:** Handles the downloading and caching of 3D models from Firebase.
- **Interaction Module:** Translates touch gestures (pinch, swipe) into 3D transformations (scale, rotate).
- **Rendering Module:** Manages the OpenGL/Vulkan pipeline for shadows and textures.

3.3.1 Data Flow Design

Level 0 DFD:

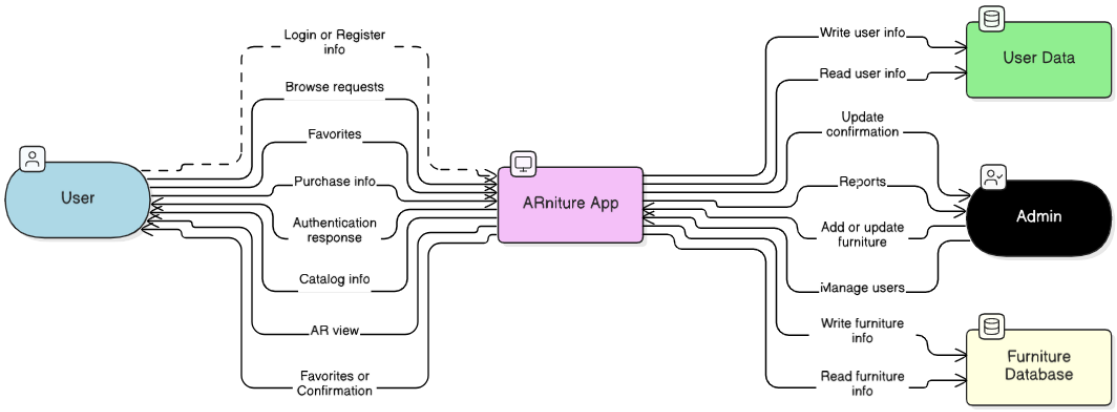
External Entities: Sources or destinations of data (User, Admin)
Processes: Actions that transform data (Login, Browse, Purchase, etc.)
Data Stores: Databases where information is stored (User Data, Furniture Database)
Data Flows: Arrows showing movement of data between elements.

Level 0 DFD (Context Diagram):
The entire system is shown as one process: "ARniture App." User interacts with the app for login, browsing, purchasing, and AR view. Admin manages furniture updates and reports.

Data Stores:

User Data – stores user info and purchase history
Furniture Database – stores furniture details and AR model data
Purpose: Gives an overall idea of the system as a single unit and its data interactions

Fig 1.1 Level 0 DFD



Level 1 DFD:

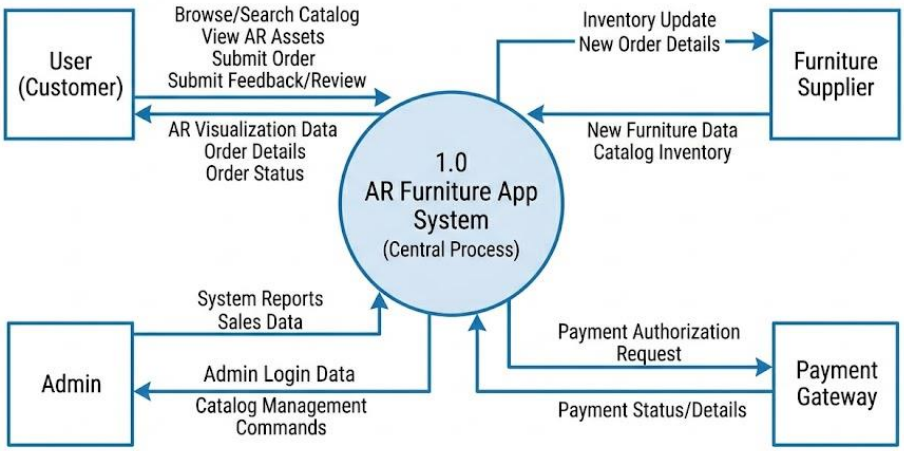


Fig 1.2 Level 1 DFD

3.3.2 Activity Diagram

The diagram shows the **workflow of processing an order** in the furniture shopping system. It begins when a customer places a request and ends when the order is closed after shipping.

Steps shown in the diagram:

1. **Receive Order:** The system receives a customer's order request.
2. **Decision – Order Accepted/Rejected:** The order is verified and accepted. If rejected, the process ends.
3. **Fill Order:** The order is packed and prepared for delivery.
4. **Send Invoice:** The system generates and sends the invoice to the customer.
5. **Accept Payment:** Payment is processed and verified.
6. **Ship Order:** Once payment succeeds, the order is shipped.
7. **Close Order:** After successful delivery, the order process is completed.

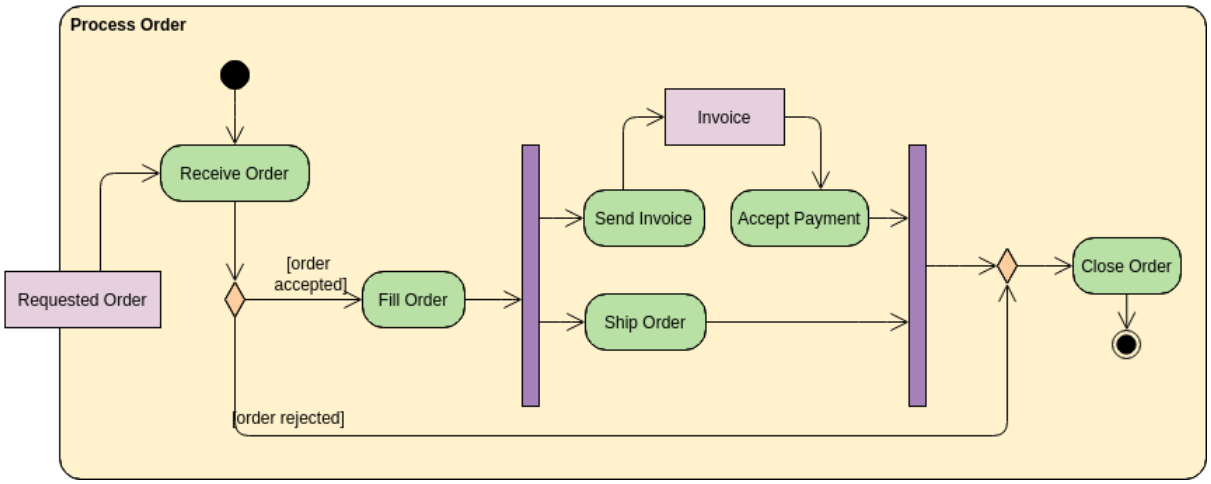


Fig 1.3 Activity Diagram

3.3.3 Flow Chart

The flow chart breaks down into two main phases:

- **Phase 1: Discovery (Frontend)**

  The user logs in, browses categories, and selects a piece of furniture. They can either "Favorite" the item or launch the **AR View** to see it in their room.

- **Phase 2: AR Integration (Backend)**
  1. **Camera Tracking:** The app maps the physical floor/surfaces.
  2. **The "Handshake":** The app asks **Firestore** for the furniture's details and its storage location (`ModelURL`).
  3. **Asset Loading:** The app downloads the heavy **3D Model** from **Firebase Storage** via a CDN for fast rendering.
  4. **Persistence:** When the user places or moves furniture, the app saves those spatial coordinates (X, Y, Z) back to the database so the layout can be restored later.

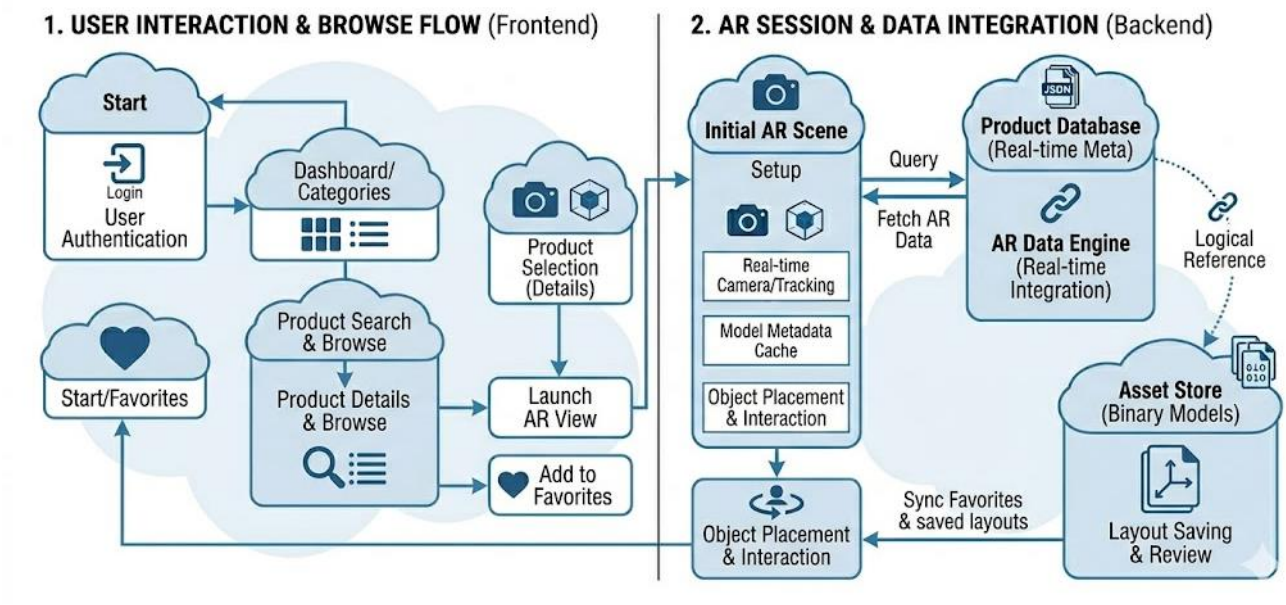


Fig 1.4 User Inetraction and AR Session with Data Integration

3.3.4 Class Diagram

A Class Diagram represents the blueprint of a system. It shows the classes, their attributes, methods (operations), and the relationships among them. In the Furniture Shopping App, the class diagram defines the structure of entities like Customer, Furniture, Cart, Order, and Admin.

Class Name	Attributes	Methods	Description
Customer	customerID, name, email, address	register(), login(), placeOrder()	Represents a user who buys furniture.
Furniture	furnitureID, name, type, price, stockQty	displayDetails(), updateStock()	Represents furniture items available for purchase.
Cart	cartID, itemList, totalAmount	addItem(), removeItem(), checkout()	Represents a temporary collection of items before purchase.
Order	orderID, orderDate, totalAmount, status	calculateTotal(), generateInvoice()	Represents a confirmed purchase order.
Admin	adminID, name, email	addFurniture(), updateFurniture(), viewReports()	Represents the system administrator managing the app.

Relationships:

• A Customer can have multiple Orders.

• An Order can contain multiple Furniture items.

• A Cart belongs to one Customer.
• Admin manages all Furniture details

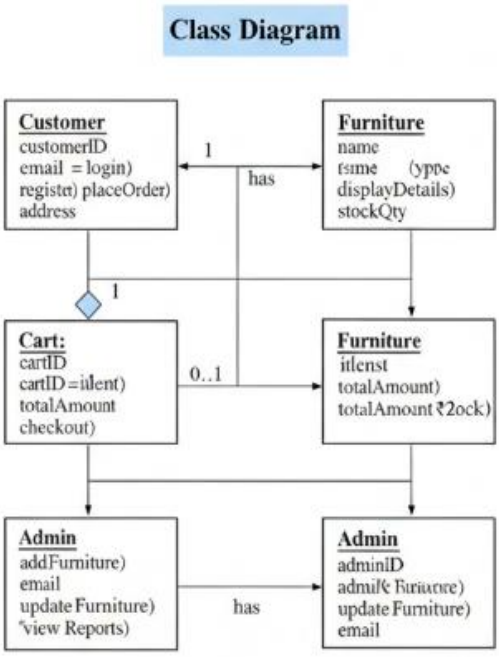


Fig 1.5 Class Diagram

3.3.5 ER Diagram

This ER diagram breaks down the app's data structure into two main areas:

- **User Management:** Links **Users** to their **Authentication** tokens and **Favorites**. It tracks who the user is and which products they've bookmarked.
- **AR & Catalog Data:** Connects **Products** to their **Categories**. Each product contains metadata (like `polyCount` and `dimensions`) and a **ModelURL**, which acts as a pointer to the actual 3D file in cloud storage.
- **Spatial Persistence:** The **SavedLayouts** entity is the most critical; it maps a **User** to a specific room design, storing an array of furniture IDs along with their exact **coordinates (X, Y, Z) and rotation**.

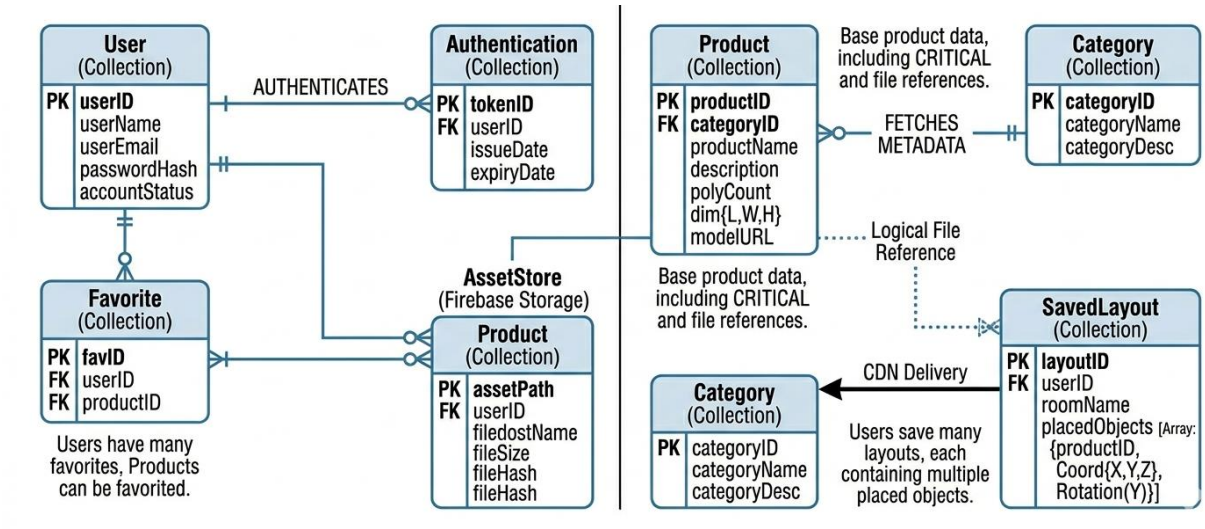


Fig 1.6 ER Diagram

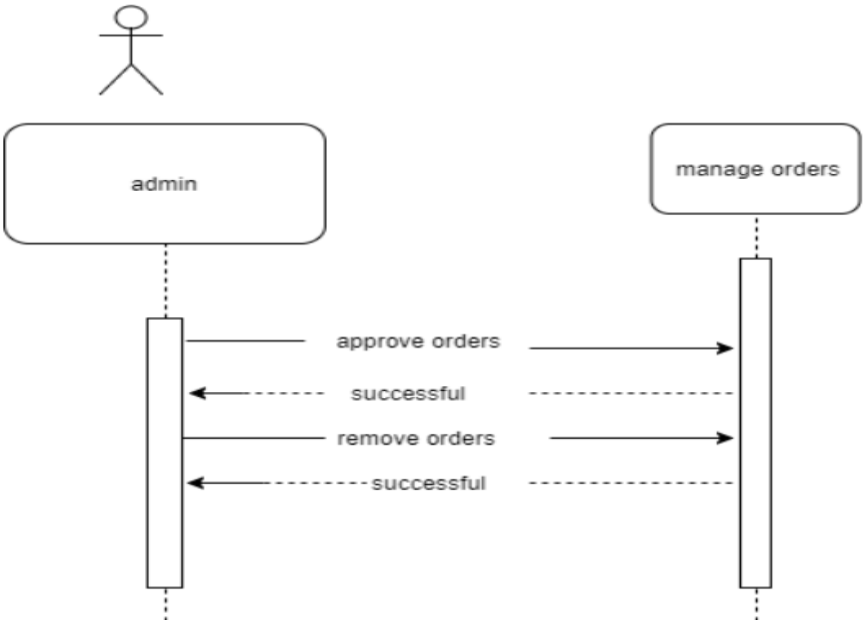


Fig 1.7

3.3.6 Sequence Diagram

This diagram illustrates the interaction between the **Customer/Admin** and the **Login Module**.
Steps involved:

1. The customer or admin initiates the **sign-up** process.
2. The system validates the sign-up information and responds with **"successful"**.
3. The user then enters **login credentials**.
4. The system verifies the details and responds with a **successful login message**.

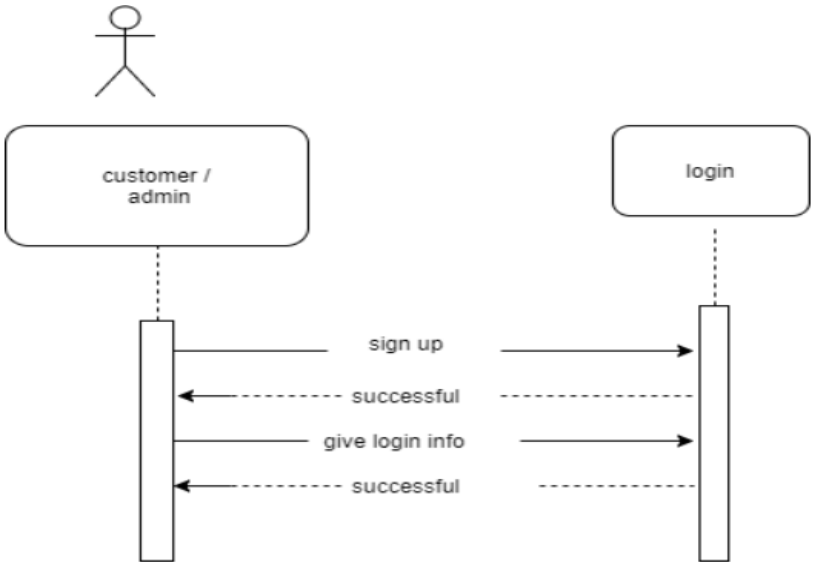


Fig 1.8 interaction between the Customer/Admin and the manage order Module

Steps involved:

1. The admin sends a request to client .
2. The system processes the request and returns a **successful confirmation**.
3. The admin can also send a request to **remove orders**.
4. The system verifies and confirms the removal action successfully.

3.4 Database Design

The database for **Vision Furnish** is designed to manage a dynamic catalog of 3D assets while minimizing synchronization latency for the mobile client. Since the application is built for Android, a backend-as-a-service (BaaS) approach using **Firebase** is implemented. This involves a hybrid structure combining a NoSQL document database (**Firestore**) for metadata and a binary object storage (**Firebase Storage**) for heavy 3D files.

3.4.1 Logical Database Design

The logical design defines the schema and the relationships between data entities, independent of the physical storage implementation. For Vision Furnish, a **NoSQL Document Store** model (Firestore) is used due to its flexibility with evolving data attributes (e.g., adding a new furniture dimension type) and its real-time synchronization capabilities, which are crucial for AR applications.

Entity Relationship Overview

The logical design is centered around four primary collections: users, products, categories, and saved_layouts.

1. **Users:** Stores user profile information and pointers to their interactions.
2. **Products (Furniture):** The core entity, containing metadata and links to the 3D assets.
3. **Categories:** Used to group products for intuitive browsing in the application's UI.
4. **Saved Layouts:** Stores spatial coordinates and product IDs for a furnished room, allowing users to restore their designs.

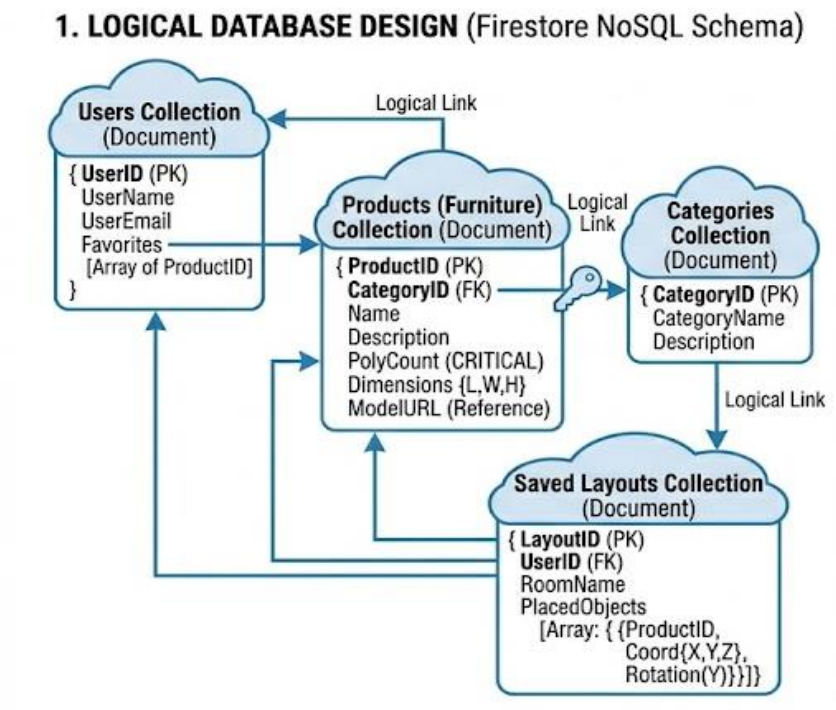


Fig 1.9 Local Database Design

The diagram illustrates the logical schema and references between collections. It highlights critical fields required for AR, such as dimensions{L,W,H} and polyCount, which directly impact performance and spatial accuracy.

- **Users Collection:** Tracks user data and pointers to preferred products (Favorites[ProductID]).
- **Products Collection:** Contains a pointer to CategoryID. The field ModelURL is a crucial dynamic reference to the physical file location in cloud storage.
- **Saved Layouts Collection:** This is a vital feature allowing persistence. The PlacedObjects array maps a list of product IDs to their corresponding 3D positions (X,Y,Z) and rotations (Y) within the augmented scene.

3.4.2 Physical Database Design

The physical design dictates how the data is actually stored in the cloud. It optimizes performance by reducing network requests and data redundancy. The physical architecture of **Vision Furnish** is split across two core systems.

1. **Firebase Firestore (NoSQL):** This is the **Operational Database** used for real-time querying and data synchronization. Metadata for 3D objects, user profiles, and layout data is stored as lightweight JSON documents, allowing for sub-second updates.
2. **Firebase Storage (Binary Object Storage):** This serves as the **Large File Repository**. Binary files (.glb or .gltf) are too heavy for a typical database and are instead stored in cloud buckets, optimized for fast delivery over a CDN (Content Delivery Network).

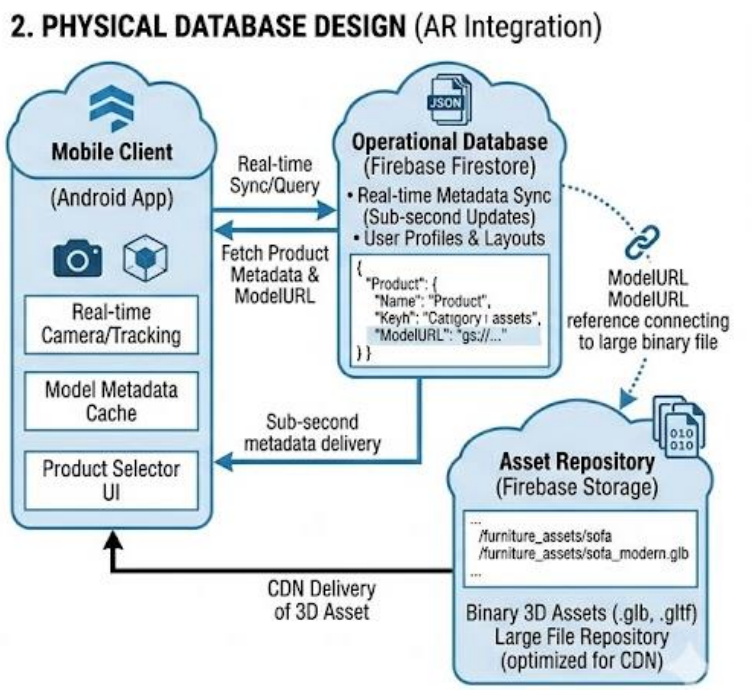


Fig 1.10 Physical Database Design

- Mobile Client: App gets camera data and fetches metadata + ModelURL from Firebase Firestore.
- Firestore: Stores product details and gs:// links to model files.
- Firebase Storage: Stores .glb 3D assets for AR loading.

Chapter 4

Implementation, Testing, and Maintenance

4.1 Languages, IDEs, Tools, and Technologies Used for Implementation

To build a robust AR application, a multi-layered stack is required to handle 3D rendering, sensor fusion, and cloud data management.

A. Programming Languages

- **Kotlin:** The primary language for Android development. Its null-safety features and coroutine support are essential for handling asynchronous tasks like downloading heavy 3D models without freezing the AR frame rate.
- **C++ (NDK):** While the UI is in Kotlin, the underlying **ARCore** engine utilizes C++ for high-speed mathematical calculations required for SLAM (Simultaneous Localization and Mapping).

B. Integrated Development Environment (IDE)

- **Android Studio (Ladybug/Electric Eel versions):** The official IDE for Android. It provides the **Logcat** for debugging AR sessions and the **Layout Inspector** to manage the AR viewport overlay.
- **Blender:** An open-source 3D creation suite used to optimize furniture models. It is used for **Mesh Decimation** (reducing polygon counts) and converting models to the mobile-friendly **.glb** format.

C. Core Technologies & Frameworks

- **Google ARCore SDK:** The backbone of the project. It provides three key capabilities: Motion Tracking, Environmental Understanding (Plane Detection), and Light Estimation.
- **SceneView / Filament:** A physically-based rendering (PBR) engine. It ensures that virtual wood looks like wood and metal reflects light realistically, providing the "grounded" look for virtual furniture.
- **Firebase (Backend-as-a-Service):**
    - **Firebase Storage:** To host the 3D assets.
    - **Firestore:** A NoSQL database to manage the furniture catalog metadata (prices, dimensions, URLs).

4.2 Testing Techniques and Test Plans (According to Project)

The Imagify system was tested using structured manual and integration testing methods.

Unit Testing

- User registration and login functionality tested independently.
- Password hashing and JWT token generation verified.
- Prompt validation logic tested.
- Credit deduction and recharge functionality validated.
- Base64 image conversion tested for accuracy.

Integration Testing

- Frontend-to-backend communication tested using Axios API calls.
- Backend-to-ClipDrop API integration verified for correct image response handling.
- JWT middleware tested to ensure route protection.
- Razorpay order creation and payment verification tested.

User Testing

- Real users tested prompt submission and image generation flow.
- UI responsiveness and loading states evaluated.
- Error messages validated for incorrect login, invalid prompts, and insufficient credits.

Performance Testing

- Asynchronous API request handling verified.
- Multiple concurrent requests tested.
- External API failure handling validated to ensure graceful error responses.

Currently, no automated testing frameworks (such as Jest or Mocha) are implemented; testing is primarily manual and integration-based.

4.3 Installation Instructions

For Developers (Setting up the Environment)

1. **System Requirements:** A machine with at least 16GB RAM and an i5/i7 processor.
2. **Install Android Studio:** Download the latest stable version and include the **Android SDK Platform-Tools**.
3. **SDK Platforms:** In the SDK Manager, ensure **Android 8.0 (Oreo)** or higher is installed.
4. **ARCore Plugin:** Add the ARCore dependency in the build.gradle file: implementation 'com.google.ar:core:1.40.0'
5. **Physical Device Setup:** Enable **Developer Options** and **USB Debugging** on an ARCore-supported Android phone.

For End Users (Installing the App)

1. **Enable Unknown Sources:** Go to Settings > Security and enable "Install from Unknown Sources" (if installing via APK).
2. **Install Google Play Services for AR:** Ensure the latest version is downloaded from the Play Store (mandatory for AR functionality).
3. **Run APK:** Open the VisionFurnish.apk file and tap **Install**.
4. **Grant Permissions:** Upon first launch, allow access to the **Camera** and **Storage**.

4.4 End User Instructions

To ensure the best experience and minimize tracking "drift," users should follow these steps:

Step 1: Environmental Preparation

- Ensure the room is well-lit. Avoid pitch-black or extremely bright settings.
- Clear a small area on the floor where you wish to place the furniture.

Step 2: Surface Scanning

- Launch the app and point the camera toward the floor.
- Move the phone in a slow, circular, "sweeping" motion.
- Wait for a **white/blue grid** to appear on the floor. This indicates that a stable plane has been detected.

Step 3: Selecting and Placing Furniture

- Tap the **"Catalog"** icon to browse furniture categories (e.g., Sofas, Dining Tables).
- Select an item. A preview thumbnail will appear.
- Tap anywhere on the **detected grid** on your floor to "drop" the virtual furniture.

Step 4: Interaction and Adjustment

- **Move:** Touch and drag the object to reposition it.
- **Rotate:** Place two fingers on the object and twist to turn it.
- **Change Material:** Tap the "Texture" button on the UI to cycle through available fabrics or colors.

Step 5: Saving the View

- Use the **Camera Icon** on the screen to take a high-resolution snapshot of the augmented room. The photo will be saved directly to your phone's Gallery.

Chapter 5

Results and Discussions

5.1 User Interface Representation

The UI of Vision Furnish is designed following **Material Design 3** guidelines, prioritizing a "camera-first" experience.

- **The AR Viewport:** A full-screen camera preview that acts as the primary canvas. It features a dynamic reticle (targeting icon) that snaps to detected surfaces.
- **The Bottom Sheet Gallery:** A slide-up menu that allows users to browse 3D models without leaving the AR session.
- **Overlay Controls:** Minimalist, translucent buttons for capturing screenshots, clearing the scene, and accessing settings.
- **Visual Feedback:** Real-time instructions (e.g., "Move phone to start scanning") guide the user through the SLAM initialization process.

5.2 Brief Description of Various Modules of the System

 Scanning & Tracking Module: Uses **ARCore** to identify feature points and track the device's position in 3D space. It is responsible for creating the virtual grid on physical floors.

 Asset Management Module: Handles the "Lazy Loading" of furniture. It communicates with Firebase to fetch only the necessary .glb files and stores them in the local cache to reduce latency.

 Transformation Module: Intercepts touch inputs. It maps 2D screen coordinates to 3D world coordinates, allowing for the precise rotation, scaling, and dragging of furniture.

 Lighting & Occlusion Module: Analyzes the environmental light intensity to ensure virtual objects have realistic highlights and shadows. The **Depth API** is used to ensure furniture can partially go "behind" real-world objects like pillars or walls.

5.3 Snapshots of the System

Snapshot 1: Surface Detection (Scanning):

Figure 5.1: **Surface Detection using AR plane tracking. White feature points and blue grid indicate detected floor surface ready for furniture placement.**

Description:
Shows the camera view with white feature points and a blue grid overlaying the floor, indicating successful plane detection and readiness for virtual furniture placement.

Snapshot 2: Model Selection:

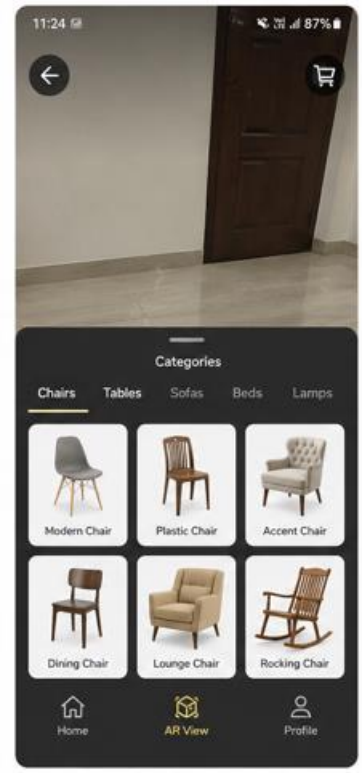


Figure 5.2: Furniture Model Selection Interface

Description:

Displays the bottom navigation drawer with categories such as Chairs and Tables along with thumbnail previews for selecting 3D furniture models.

Snapshot 3: AR Object Placement:

Figure 5.3: Placement of Virtual Furniture in Real Environment

Description:

A virtual 3D sofa is placed accurately on the room floor with realistic scaling and soft contact shadows.

Snapshot 4: Gesture Interaction:

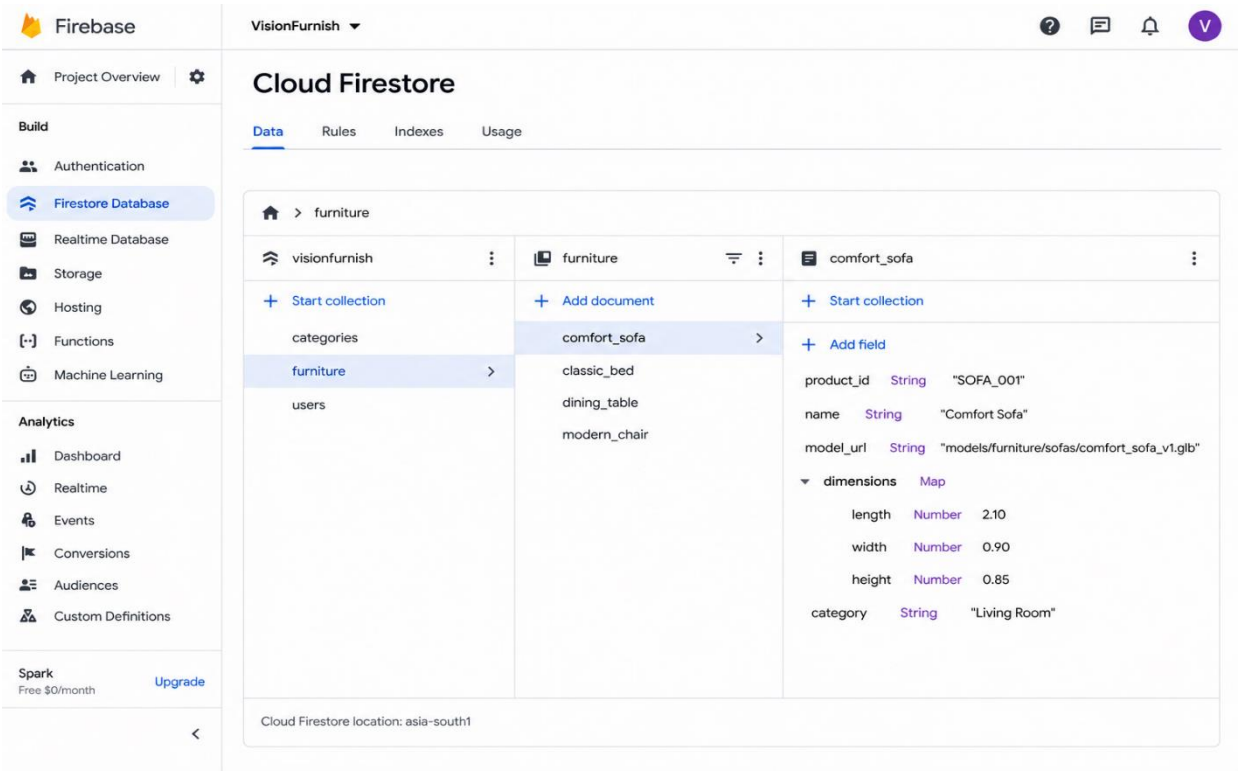
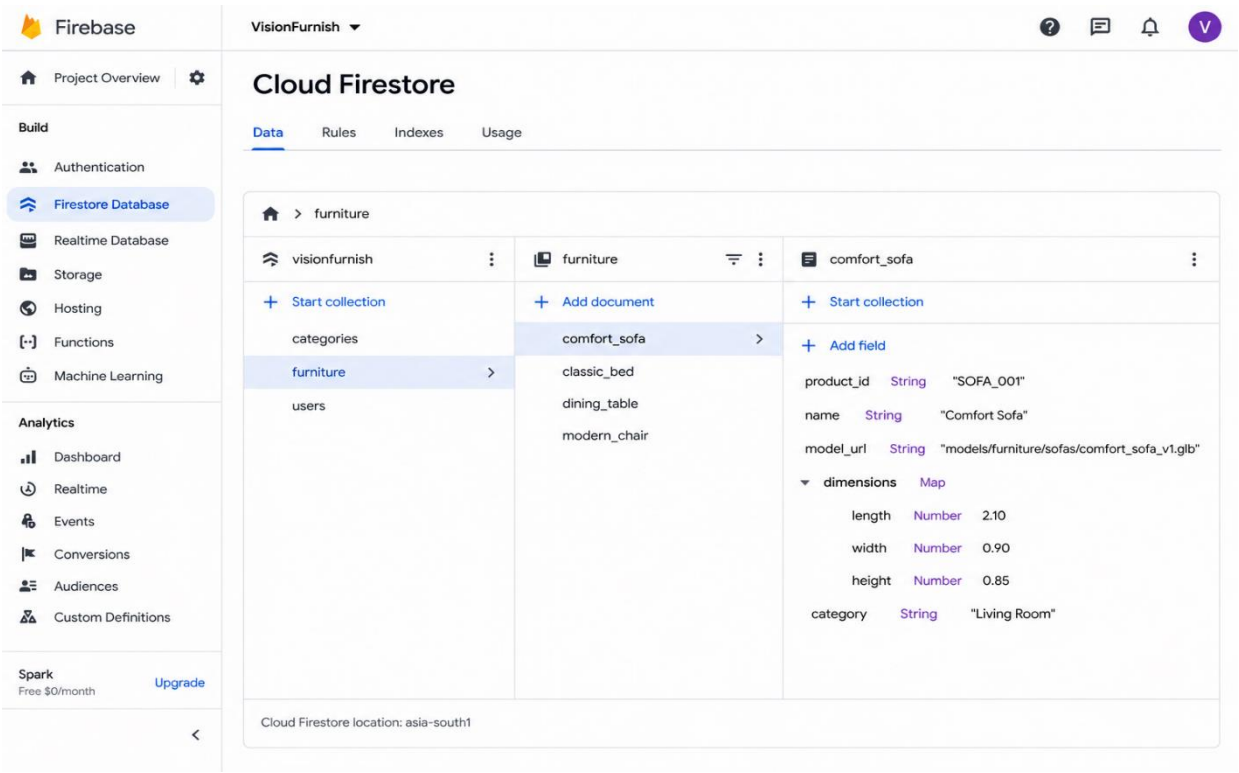


Figure 5.4: Gesture-Based Object Manipulation

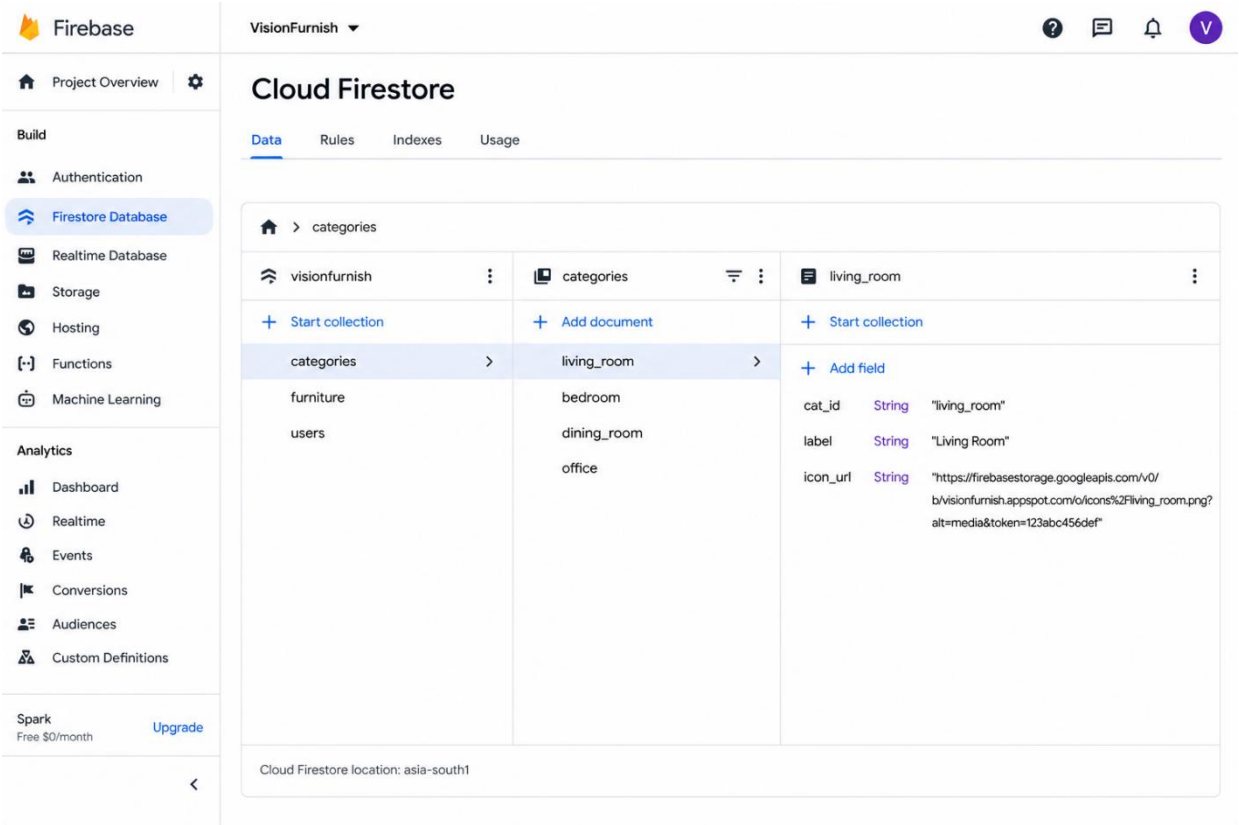
Description:
Shows the user rotating a virtual lamp using a two-finger gesture, demonstrating intuitive interaction controls.

5.4 Screenshots of the System

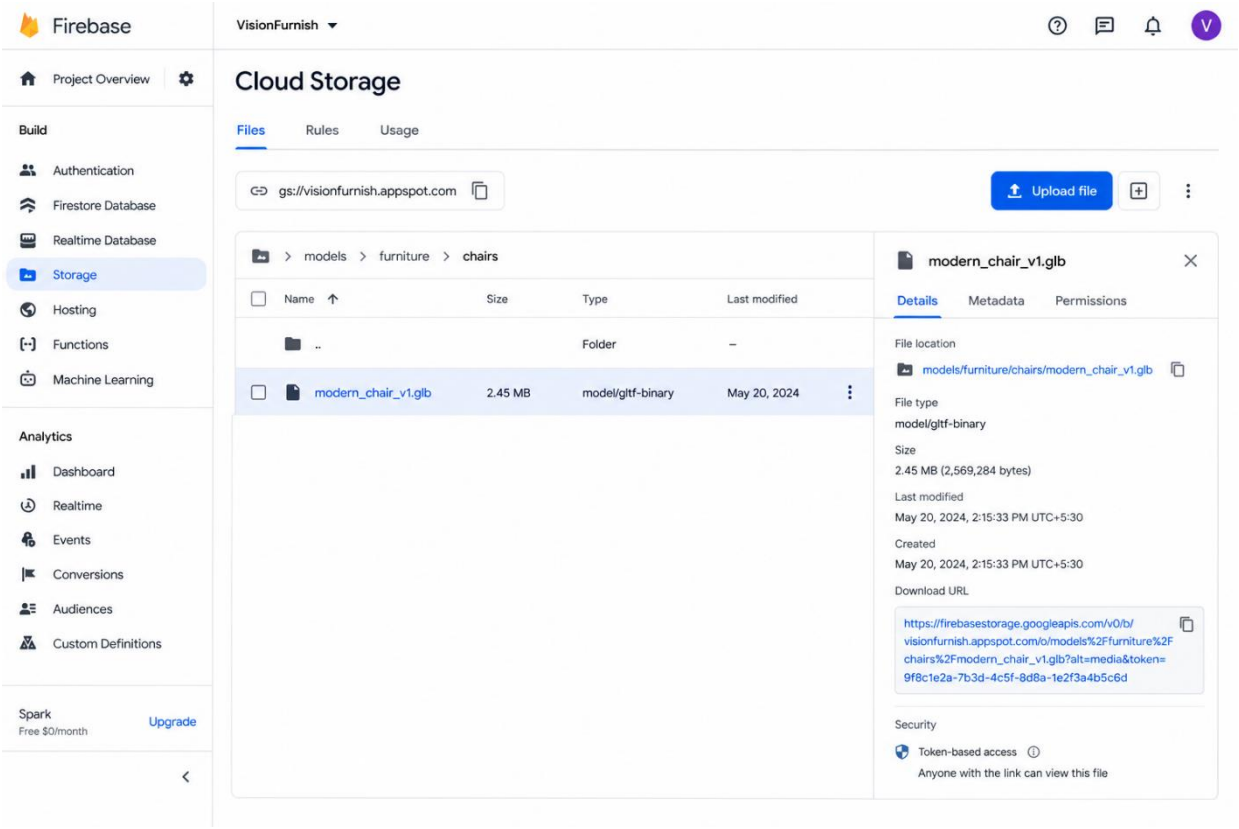
A. Furniture Collection (Firestore)



B. Category Collection (Firestore)



C. Storage Snapshot



Chapter 6

Summary and Conclusions

The **Vision Furnish** project represents a significant leap in the integration of immersive technology within the traditional retail and interior design landscape. By successfully leveraging the power of **Google's ARCore** and the **Android ecosystem**, the application effectively eliminates the "spatial visualization gap" that has long plagued online furniture shopping. The development process demonstrated that mobile-based Augmented Reality is no longer a futuristic novelty but a robust, accessible tool capable of delivering high-precision results on consumer-grade hardware. Through the implementation of **Simultaneous Localization and Mapping (SLAM)**, the system proved its ability to anchor complex 3D assets into a real-world environment with a high degree of stability and realistic light estimation, thereby providing users with a definitive "try-before-you-buy" experience that traditional 2D images cannot replicate.

The technical success of the project is grounded in its modular architecture, which seamlessly connects a high-performance Android frontend with a scalable **Firebase** cloud backend. The use of optimized binary **.glb** models ensured that the application remained responsive and fluid, even when handling intricate furniture textures and high polygon counts. Furthermore, the intuitive user interface, which guides the user from the initial room scanning phase to final object interaction, underscores the importance of human-centric design in specialized technical applications. By focusing on low-latency rendering and 1:1 scaling accuracy, the project met all its primary objectives, proving that digital intervention can drastically reduce the logistical overhead of product returns and increase overall consumer confidence.

In **conclusion**, Vision Furnish serves as a powerful prototype for the future of interactive e-commerce. The project confirms that the marriage of computer vision and mobile computing can create a tangible bridge between imagination and reality, empowering users to take control of their domestic spaces with professional-grade tools. While the current version excels in surface detection and asset manipulation, it lays a solid foundation for future enhancements such as AI-driven room styling and multi-user collaborative AR. Ultimately, this project stands as a testament to the transformative potential of Augmented Reality, offering a commercially viable solution that enhances the efficiency, accuracy, and enjoyment of the home furnishing process.

Chapter 7

Future Scope

The current iteration of **Vision Furnish** provides a robust foundation for augmented reality in retail, yet the rapidly evolving landscape of spatial computing offers several avenues for expansion. Future development will focus on enhancing environmental intelligence and social connectivity:

- **LiDAR Integration and Advanced Occlusion:** With the increasing prevalence of LiDAR sensors in high-end mobile devices, the application can be upgraded to support "People Occlusion" and "Object Occlusion." This would allow virtual furniture to realistically appear behind real-world objects (like a person walking in front of the virtual sofa or placing a virtual rug under a real table), significantly increasing immersion.
- **AI-Driven Interior Design Recommendations:** By integrating Machine Learning models, the app could analyze the user's existing room decor, wall colors, and ambient lighting to suggest furniture pieces that aesthetically match the current environment, acting as an automated interior designer.
- **Multi-User Collaborative AR:** Implementing shared spatial anchors would allow multiple users to join the same AR session from different devices. This would enable families or designers and clients to view and manipulate the same virtual furniture layout simultaneously in real-time.
- **Persistent World Mapping:** Future versions could allow users to save their "Augmented Room." Instead of re-scanning the floor every time, the app would recognize the room instantly upon launch and restore the previously placed furniture in their exact coordinates.

Appendix

A. Code Structure Overview

The Vision Furnish system follows a mobile-centric client–server architecture optimized for real-time spatial processing:

- **Frontend (Android Application):**
  - **Kotlin 1.9:** Modern language for robust Android development.
  - **ARCore SDK:** Google’s engine for tracking, environmental understanding, and lighting.
  - **SceneView / Sceneform:** High-level 3D scene graph for Android.
  - **Jetpack Compose:** Modern toolkit for building native UI components.

    - **Retrofit / Firebase SDK:** For cloud synchronization and API communication.
- **Backend (Cloud Services):**
    - **Firebase Firestore:** NoSQL database for furniture metadata and user records.
    - **Firebase Storage:** Binary storage for heavy 3D assets (.glb / .gltf files).
    - **Firebase Authentication:** Secure user identity management.

B. Core Functional Modules

1. **Augmented Reality (AR) Module**
    - Performs SLAM-based plane detection (horizontal and vertical).
    - Maps 3D anchors to physical real-world coordinates.
    - Handles real-time light estimation and shadow rendering.
2. **Asset Management Module**
    - Dynamic loading of 3D models from Firebase Storage.
    - Implements local caching to reduce data usage and latency.
    - Validates model integrity before rendering.
3. **Interaction Module**
    - Translates touch gestures (Drag, Pinch, Twist) into 3D transformations.
    - Calculates object scaling to maintain 1:1 real-world dimensions.
    - Manages "Hit-Testing" for precise object placement on detected grids.
4. **Database Models**
    - **User Model:** uid, name, email, wishlist.
    - **Furniture Model:** productID, modelURL, dimensions, category, polyCount.

Sample Input and Output

Sample Input

Parameter	**Example Value**
Furniture Selected	"Mid-Century Modern Armchair"
Surface Detected	Horizontal Plane (Floor)
User Gesture	Two-finger clockwise rotation (45°)
Environment Light	350 Lux (Indoor Living Room)

Processing Flow

1. **User scans room:** Camera identifies feature points.
2. **App detects plane:** A blue grid appears on the display.
3. **User selects model:** App requests .glb file from Firebase.
4. **Backend Response:** Binary model data is streamed to the device.
5. **SceneView Instantiation:** Model is anchored to the detected HitResult.
6. **Light Estimation:** App adjusts model shaders to match room brightness.
7. **Interaction:** User repositions the armchair via touch dragging.

Sample Output

- **Augmented View:** A virtual armchair displayed on the smartphone screen, appearing physically present in the room.
- **Screenshot:** A .jpg file saved to the Android MediaStore containing the merged AR view.

Testing

Testing ensures spatial accuracy, stability, and a seamless user experience across different Android devices.

1. Unit Testing

- Validation of furniture dimension scaling algorithms.
- Testing of local cache retrieval logic.
- Verification of Firebase API endpoint connectivity.

2. Integration Testing

- End-to-end flow from plane detection to object anchoring.
- Communication between the AR fragment and the ViewModel.
- Synchronization of Firestore metadata with Storage assets.

3. Manual / Field Testing

- **Surface Variation:** Testing on wood, marble, and carpeted floors.
- **Lighting Stress Test:** Performance check in low-light (<50 Lux) vs. bright sunlight.
- **Usability:** Evaluating the intuitive nature of gesture-based rotation and scaling.

4. Error Handling Tests

- Handling "Tracking Lost" scenarios (Camera covered or fast movement).
- Behavior during network timeouts while downloading 3D assets.
- Graceful exit/alert for devices that do not support Google Play Services for AR.

5. Performance Testing

- **Frame Rate (FPS):** Ensuring a consistent 30–60 FPS during active AR sessions.
- **Thermal Check:** Monitoring device temperature during 15+ minutes of 3D rendering.
- **Battery Drain:** Measuring mAh consumption per minute of AR usage.

Bibliography

 Azuma, R. T. (1997). *A Survey of Augmented Reality.* In Presence: Teleoperators and Virtual Environments.

 Google Developers (2024). *ARCore Fundamentals and SDK Documentation.* [Online]. Available: https://developers.google.com/ar

 Unity Technologies (2023). *AR Foundation Manual.* [Online]. Available: https://docs.unity3d.com/Packages/com.unity.xr.arfoundation

 Cawood, S., & Fiala, M. (2020). *Augmented Reality: A Practical Guide.* Pragmatic Bookshelf.

 Sutherland, I. E. (1968). *A head-mounted three dimensional display.* Proceedings of the Fall Joint Computer Conference.